SETUP & PROMPT GUIDE

Hermes Agent Team: AI Content Pipeline with Supabase + X Algorithm Optimizer

A step-by-step guide to building an autonomous multi-agent content pipeline using Hermes Kanban, TinyFish web search, Supabase persistent storage, and open-source X algorithm virality scoring.

Hermes Version: v0.13.0 (v2026.5.7) **Feature:** Kanban Multi-Agent Coordination

Pipeline: Research → Script → X Optimizer → Supabase

CONTENTS

1. Overview & Architecture
2. Prerequisites & Accounts
3. Step 1 — Deploy Hermes on Railway
4. Step 2 — Spin Up the Kanban Agent Team
5. Step 3 — Configure Supabase Pipeline Schema
6. Step 4 — Research Agent (TinyFish Topic Finder)
7. Step 5 — Script Agent
8. Step 6 — X Algorithm Optimizer Agent
9. Step 7 — Full Pipeline Run
10. Bonus — Weekly Cron Schedule
11. All Prompts Reference Table

Overview & Architecture

This guide walks through building a fully autonomous AI content pipeline using Hermes v0.13.0's new Kanban multi-agent coordination feature. Instead of one agent doing everything sequentially, the Kanban board lets you run a coordinated team of specialist agents in parallel, each owning a single stage of the pipeline.

What Gets Built

- **Research Agent** — uses TinyFish web search to find trending YouTube topics and score each for virality potential, saving results to Supabase.
- **Script Agent** — reads the top-rated research topic and writes a full structured video script, saving the draft to Supabase.
- **X Optimizer Agent** — applies open-sourced X algorithm engagement weights to rewrite the script as a high-reach post and thread, saving the output with full signal metadata to Supabase.
- **Storage Agent** — coordinates Supabase writes across all pipeline stages.

How the Agents Coordinate

The Kanban board is the shared communication layer. Agents do not call each other directly — they read from and write to the board and to Supabase. When an upstream agent moves its card to Done, that event unblocks the next downstream agent. This means Script and X Optimizer agents can run in parallel across multiple topics simultaneously.

Supabase as Pipeline Memory

Supabase provides the persistent state for the entire pipeline. Four tables store each stage of output:

Table	Purpose	Key Columns
`topics`	Research Agent output	topic title, virality_potential (1–10), source, created_at
`scripts`	Script Agent drafts	topic_id (FK), draft_text, status (draft/approved)
`x_posts`	X Optimizer output	script_id (FK), post_text, thread_text, virality_score, signals_applied (JSONB)
`pipeline_runs`	Kanban state tracking	run_id, stage, status, agent_id, started_at, completed_at

Free Tier Note: The Supabase free tier is permanent, $0/month, includes a dedicated Postgres instance, 500 MB storage, and unlimited API requests. This project comfortably fits within those limits — no paid plan required.

Prerequisites & Accounts

You will need accounts and API keys for three services before starting. All free tiers are sufficient.

Service	What You Need	Cost	Where to Get It
Railway	Account to deploy Hermes container	Free tier available	railway.com
DeepSeek	API key — the LLM powering Hermes agents	Pay per use (low cost)	platform.deepseek.com
Supabase	Project URL + Service Role Key	Free tier (permanent)	supabase.com → Settings → API
TinyFish	API key for web search and fetch	Free (search + fetch)	agent.tinyfish.ai

Before you start: Copy your Supabase Project URL and Service Role Key from **Supabase Dashboard → Settings → API**. You will need both when installing the Supabase skill in Step 3. Also have your TinyFish API key ready for Step 4.

GitHub Repo

All prompts from this guide are available in the free GitHub repository:

`github.com/derekcheungsa/ai-automation-resources`

Steps reference specific files (e.g. `prompts/01-spin-up-agents.md`). You can copy prompts directly from there rather than typing them manually.

1 Deploy Hermes on Railway

Time in video: 1:15–2:15

Hermes is deployed via a Railway one-click template. This step gets the agent environment running and exposes the new Kanban feature.

Steps

1. Go to: `railway.com/deploy/hermes-agent-by-no-1`
2. Click **Deploy**.
3. In the Railway dashboard, open the **environment variables** panel.
4. Add your **DeepSeek API key** as an environment variable.
5. Wait approximately 2 minutes for Railway to boot the Hermes container.
6. Open the Hermes UI. Confirm the **Kanban tab** is visible in the left sidebar — this is new in v2026.5.7.

> **First time with Hermes?** The YouTube channel description links to previous setup walkthrough videos if you need a more detailed environment configuration guide.

2 Spin Up the Kanban Agent Team

Time in video: 2:15–4:30

The Kanban board uses six columns by default: **Triage | Todo | Ready | Running | Blocked | Done**. Agents pick up work automatically and update card status as they progress. Nothing gets dropped, nothing runs twice, and the board shows exactly where everything stands at any moment.

How Kanban Multi-Agent Works

- Each agent is assigned a specific card and profile — it focuses only on its own task.
- Agents run in parallel; they do not share state directly.
- Coordination happens through the board (card status) and Supabase (shared data).
- Each card has a comment thread — agents log their reasoning, queries, and decisions there, which downstream agents can read.

Prompt — Create the Four-Agent Team

Copy this prompt from `prompts/01-spin-up-agents.md` in the GitHub repo and paste it into Hermes chat:

PROMPT 1 — SPIN UP AGENT TEAM

```
I'm building a content creation pipeline. Create four Kanban tasks for
these agents:

(1) Research Agent — find trending YouTube topics using TinyFish web
search and save results to Supabase.
(2) Script Agent — turn a research topic into a complete video script.
(3) X Optimizer Agent — apply X algorithm virality signals to generate an
optimized post and thread.
(4) Storage Agent — coordinate Supabase writes across all stages.

Assign each to its own profile and set them as Ready.
```

What to Expect

- Four task cards appear in the **Ready** column with individual profile assignments.
- Nudge the dispatcher to skip the default 60-second wait — cards cascade to **Running**.
- All four cards move to Running simultaneously. Each agent has its own process, tools, and context.

Card comment threads: After an agent completes its work, click into its Done card to see the per-task comment thread. The agent writes its reasoning, queries run, and scoring logic directly there. Downstream agents read this as context.

3 Configure Supabase Pipeline Schema

Time in video: 4:30–6:30

The Supabase skill lets you build and manage database schema entirely through natural language — no SQL Editor required. You will install the skill and then ask Hermes to create all four tables.

Install the Supabase Skill

Paste this into Hermes chat, replacing the placeholders with your actual values from **Supabase → Settings → API**:

PROMPT 2 — INSTALL SUPABASE SKILL

```
Install the Supabase agent skill. My Supabase project URL is
[YOUR_PROJECT_URL] and my service role key is [YOUR_SERVICE_ROLE_KEY].
```

Hermes installs the skill and confirms the connection to your Supabase project.

Create the Pipeline Schema

The full prompt is in the GitHub repo. Here is the complete version to paste:

PROMPT 3 — BUILD PIPELINE SCHEMA

```
Create four tables in my Supabase project for an AI content pipeline:

1. topics — stores research results with fields: id, topic_title,
virality_potential (integer 1-10), source_url, notes, created_at.
2. scripts — stores video script drafts with fields: id, topic_id (FK →
topics.id), draft_text, status (default: 'draft'), created_at,
updated_at.
3. x_posts — stores X-optimized content with fields: id, script_id (FK →
scripts.id), post_text, thread_text, virality_score (integer 1-100),
signals_applied (JSONB), created_at.
4. pipeline_runs — tracks Kanban coordination state with fields: id,
run_id, stage, agent_id, status, started_at, completed_at.

Add foreign keys linking each stage. Add RLS policies so each agent
writes only to the tables it owns. Confirm once all tables and policies
are created.
```

What to Expect

- Watch the Supabase dashboard — tables appear in real time: `topics`, `scripts`, `x_posts`, `pipeline_runs`.
- RLS policies are added automatically per the prompt specification.
- The `signals_applied` column in `x_posts` is JSONB — this stores the exact X algorithm weights the optimizer applied (reply multipliers, link penalty, timing data).

Why JSONB for signals_applied? The X algorithm produces a variable set of weighted signals per post. JSONB lets the agent write the full signal breakdown as structured data without requiring a fixed schema. You can query specific signal keys later using Supabase's JSONB operators.

4 Research Agent — TinyFish Topic Finder

Time in video: 6:30–9:00

TinyFish is a web infrastructure platform purpose-built for AI agents. It provides search, fetch, browser, and agent capabilities under a single API key. Web search and fetch are completely free.

Install the TinyFish Skill

Sign up at `agent.tinyfish.ai` to get your API key. Then install the skill into Hermes:

PROMPT 4 — INSTALL TINYFISH SKILL

```
Please install the TinyFish skill. My TinyFish API key is
[YOUR_TINYFISH_API_KEY].
```

The skill teaches Hermes when to reach for TinyFish search versus fetch, and how to call each endpoint correctly.

Configure the Research Agent

PROMPT 5 — CONFIGURE RESEARCH AGENT

```
Configure the Research Agent with the following task:

Use TinyFish to search the web and find five trending AI automation
topics targeted at non-technical audiences. For each topic, score it 1–10
for virality potential based on search volume signals and engagement
indicators. Save all results to the Supabase topics table with the topic
title, virality score, and source. Mark your Kanban card Done when
finished.
```

What to Expect

- The Research Agent runs live TinyFish search queries — results come from real web data, not model guesses.
- The `topics` table in Supabase fills with rows as the agent writes each result.

- Example top result: *"AI Agents for Beginners: No-Code Build Tutorials 2026"* — trending score 95.
- After completion, click the Done card to read the agent's comment thread: queries run, scoring rationale, and why specific topics ranked higher.

TinyFish web search is free: Search and fetch endpoints have no cost per call on the free tier. This makes the Research Agent the lowest-cost component of the pipeline.

5 Script Agent — Turning Research Into a Draft

Time in video: 9:00–10:15

The Script Agent reads the Research Agent's Supabase output, selects the top-scored topic, and writes a complete structured video script. It saves the draft to Supabase in real time as it writes.

Configure the Script Agent

PROMPT 6 — CONFIGURE SCRIPT AGENT

```
Configure the Script Agent with the following task:

Read the Supabase topics table and pick the topic with the highest
virality_potential score. Write a full video script for that topic
structured as: cold open, seven steps, payoff section, friction/challenge
moment, and outro. Save the complete script to the Supabase scripts table
as a draft (status: 'draft'). Mark your Kanban card Done when the script
is saved.
```

What to Expect

- The Script Agent reads the top topic from Supabase (e.g. "AI Agents for Beginners: No-Code Build Tutorials 2026", score 95).
- The `draft_text` column in the `scripts` table fills in real time as the agent writes.
- Status is set to `draft` on save — ready for the X Optimizer to pick up.

6 X Algorithm Optimizer Agent

Time in video: 10:15–12:30

The X Optimizer Agent applies verified, open-sourced X algorithm engagement weights to rewrite the script draft as a post and thread engineered for maximum algorithmic reach.

Key Algorithm Signals (from Public Source Code)

Signal	Weight / Rule	Source
Replies	27x the value of a like	Twitter/the-algorithm (April 2023 release)
Author replies to commenters	150x the value of a like	Twitter/the-algorithm (April 2023 release)
External links in root post	Suppressed/penalized since 2023	X public confirmation (2024)
Early velocity	Engagement in first 30–60 min is a top ranking signal	Twitter/the-algorithm (April 2023 release)
Optimal post timing	7–9 am or 6–8 pm for maximum initial reach	Derived from velocity signal

Algorithm history: X first open-sourced their recommendation algorithm in April 2023. A Grok-based rewrite (Phoenix) was released in January 2026. The 2023 engagement weights remain publicly available in the `twitter/the-algorithm` GitHub repository.

Configure the X Optimizer Agent

PROMPT 7 — CONFIGURE X OPTIMIZER AGENT

```
Configure the X Optimizer Agent with the following task:

Read the latest script draft from the Supabase scripts table (status:
'draft'). Rewrite it as a high-reach X post and thread using these
algorithm rules:
- Engineer for replies (replies are worth 27x a like in the ranking
algorithm)
- Do NOT include external links in the root post — put any links in the
first reply
- Target 5 or more replies in the first 15 minutes (early velocity
signal)
- Recommend posting at 7–9am or 6–8pm for maximum initial reach
- Use a specific claim with an honest caveat to invite responses

Score the output 1–100 for virality. Save to the Supabase x_posts table
with post_text, thread_text, virality_score, and a signals_applied JSONB
field documenting which algorithm weights were applied and how. Mark your
Kanban card Done when finished.
```

What to Expect in Supabase

After the agent completes, expand the `signals_applied` JSONB field in the `x_posts` table. You should see entries similar to:

- **reply_bait_structure** — confirms the post is structured to generate reply responses
- **no_external_link_root** — link placed in first reply, not root post
- **early_velocity_target** — structured for 5+ replies within 15 minutes
- **post_timing** — recommended window (7–9am or 6–8pm)

Example Output

Root post (virality score: 83):

EXAMPLE X POST OUTPUT

```
I built a no-code CRM with AI agents in 47 minutes. Hermes Agent + n8n + Supabase. No code. No SQL. No canvas. Here's what it can do (and where it broke): [thread follows]
```

Specific claim, honest caveat, thread continuation — written to generate replies because the algorithm rewards them 27x over likes.

7 Full Pipeline Run — Everything in One Prompt

Time in video: 12:30–13:45

With all agents configured, you can run the entire research-to-X-optimized pipeline from a single prompt. Hermes coordinates the agent team automatically via the Kanban board — agents do not start downstream work until upstream tasks complete.

Full Pipeline Prompt

PROMPT 8 — FULL PIPELINE RUN

```
Run the full content pipeline. Research five new AI automation topics
using TinyFish, pick the top two by virality potential, write a script
for each, optimize both for X using algorithm signals, and save all
outputs to Supabase. Coordinate via Kanban — agents should not start
downstream work until upstream tasks complete.
```

Kanban Execution Sequence

1. **Research Agent runs first** — finds and scores five topics, saves top two to Supabase. Research cards move to Done.
2. **Two Script Agents go Running simultaneously** — one per topic. Scripts written in parallel, both saved to Supabase as drafts.
3. **Two X Optimizer Agents go Running simultaneously** — one per script. Parallel virality scoring.
4. **All cards move to Done** — full pipeline complete.

Expected Results

Supabase Table	New Rows	Notes
`topics`	2 rows	Top two topics by virality score
`scripts`	2 rows	Status: draft, full script text
`x_posts`	2 rows	Virality scores: ~81 and ~78, signals_applied JSONB populated
`pipeline_runs`	Multiple rows	One row per stage per agent run

Typical runtime: Approximately 8 minutes end-to-end for a two-topic full pipeline run with parallel script and optimization agents.

+ Bonus — Weekly Cron Schedule

Time in video: 13:45–14:45

Once the pipeline runs correctly end-to-end, you can schedule it to run automatically on a weekly cadence. Hermes creates a cron-triggered orchestrator from a single prompt.

Schedule the Pipeline

PROMPT 9 — WEEKLY CRON SCHEDULE

```
Set up a weekly content pipeline run every Monday at 7am. Use the same
full pipeline configuration — TinyFish research, top two topics, scripts,
X optimization, all saved to Supabase.
```

What Happens

- Hermes creates a cron-triggered orchestrator tied to the configured pipeline.
- Every Monday at 7am, the agent team runs automatically: Research → Script → X Optimizer → Supabase.
- All agents complete and return to idle.
- Open Supabase on Monday morning to find the week's content planned — no human in the loop for any stage.

All Prompts — Quick Reference Table

All prompts from this guide in order. Copy directly or pull from the GitHub repo (`prompts/` folder).

#	Step	Prompt (abbreviated — see full prompt in the relevant section)
1	Spin Up Agent Team	I'm building a content creation pipeline. Create four Kanban tasks for these agents: (1) Research Agent… (2) Script Agent… (3) X Optimizer Agent… (4) Storage Agent… Assign each to its own profile and set them as Ready.
2	Install Supabase Skill	Install the Supabase agent skill. My Supabase project URL is [URL] and my service role key is [KEY].
3	Build Supabase Schema	Create four tables — topics, scripts, x_posts, pipeline_runs — with foreign keys, JSONB columns for signal data, and RLS policies so each agent writes only to its own tables.
4	Install TinyFish Skill	Please install the TinyFish skill. My TinyFish API key is [KEY].
5	Research Agent Config	Configure the Research Agent: use TinyFish to find five trending AI automation topics for non-technical audiences, score each 1–10 for virality, save to topics table, mark Done when finished.
6	Script Agent Config	Configure the Script Agent: pick the top virality_potential topic from Supabase, write a full script — cold open, seven steps, payoff, friction, outro — save as draft, mark card Done.
7	X Optimizer Agent Config	Configure X Optimizer Agent: read latest script draft, rewrite as post + thread — engineer for replies (27x a like), no external links in root post, target 5+ replies in 15 min, post at 7–9am or 6–8pm. Score 1–100, save to x_posts with signals_applied JSONB, mark Done.
8	Full Pipeline Run	Run the full content pipeline. Research five new topics using TinyFish, pick top two, write a script for each, optimize both for X, save all to Supabase. Coordinate via Kanban — no downstream work until upstream tasks complete.
9	Weekly Cron Schedule	Set up a weekly content pipeline run every Monday at 7am.

Troubleshooting Quick Reference

Issue	Likely Cause	Fix
Kanban tab not visible	Hermes version below v2026.5.7	Redeploy from Railway template to get v0.13.0
Supabase skill install fails	Wrong key or URL format	Confirm URL format: `https://[project-ref].supabase.co`. Use Service Role key, not anon key.
TinyFish search returns no results	Invalid API key or skill not installed	Re-run Prompt 4 with a fresh API key from agent.tinyfish.ai
Agents stuck in Running	Downstream agent started before upstream Done	Include "Coordinate via Kanban — agents should not start downstream work until upstream tasks complete" in the pipeline prompt
x_posts virality score low (<50)	Script too generic or lacks a specific claim	Ensure the Script Agent prompt requests a specific data point or claim in the cold open
signals_applied JSONB empty	X Optimizer prompt missing JSONB instruction	Re-run Prompt 7 verbatim — it explicitly requests the JSONB field with signal breakdown